

# COER UNIVERSITY

## DBMS LAB ASSIGNMENT | LAB SHEET 3: SQL FUNCTIONS

### Section A: Number Functions

#### Question 1

*Question: Demonstrate ABS() with suitable examples.*

**Answer:**

```
SELECT ABS(-25.5) AS AbsVal, ABS(10) AS PosVal;
```

**Explanation:** ABS(x) returns the positive absolute value of a number.

#### Question 2

*Question: Demonstrate ROUND() with suitable examples.*

**Answer:**

```
SELECT ROUND(123.4567, 2) AS RoundedTwoDec, ROUND(123.4567, 0) AS RoundedInt;
```

**Explanation:** ROUND(x, d) rounds x to d decimal places.

#### Question 3

*Question: Demonstrate CEIL() and FLOOR().*

**Answer:**

```
SELECT CEIL(12.3) AS CeilVal, FLOOR(12.8) AS FloorVal;
```

**Explanation:** CEIL() returns the smallest integer  $\geq x$ ; FLOOR() returns the largest integer  $\leq x$ .

#### Question 4

*Question: Demonstrate MOD() with suitable examples.*

**Answer:**

```
SELECT MOD(17, 5) AS Remainder, 17 % 5 AS AltRemainder;
```

**Explanation:** MOD(N, M) returns the remainder of N divided by M.

#### Question 5

*Question: Calculate the rounded salary of employees.*

**Answer:**

```
SELECT Name, Salary, ROUND(Salary, 0) AS RoundedSalary FROM Employee;
```

**Explanation:** Applies ROUND() to the Salary column.

## Section B: Aggregate Functions

### Question 6

*Question: Find the total number of students using COUNT().*

**Answer:**

```
SELECT COUNT(*) AS TotalStudents FROM Student;
```

**Explanation:** COUNT(\*) returns the total number of rows in the table.

### Question 7

*Question: Find the total salary of all employees using SUM().*

**Answer:**

```
SELECT SUM(Salary) AS TotalSalary FROM Employee;
```

**Explanation:** SUM() calculates the sum of all values in a numeric column.

### Question 8

*Question: Find the average salary using AVG().*

**Answer:**

```
SELECT AVG(Salary) AS AvgSalary FROM Employee;
```

**Explanation:** AVG() calculates the mean value of a numeric column.

### Question 9

*Question: Find the highest salary using MAX().*

**Answer:**

```
SELECT MAX(Salary) AS HighestSalary FROM Employee;
```

**Explanation:** MAX() retrieves the maximum value from a column.

### Question 10

*Question: Find the lowest salary using MIN().*

**Answer:**

```
SELECT MIN(Salary) AS LowestSalary FROM Employee;
```

**Explanation:** MIN() retrieves the minimum value from a column.

## Section C: Character Functions

### Question 11

*Question: Demonstrate UPPER() on student names.*

**Answer:**

```
SELECT Name, UPPER(Name) AS UpperName FROM Student;
```

**Explanation:** UPPER() converts text to uppercase.

### Question 12

*Question: Demonstrate LOWER() on student names.*

**Answer:**

```
SELECT Name, LOWER(Name) AS LowerName FROM Student;
```

**Explanation:** LOWER() converts text to lowercase.

### Question 13

*Question: Demonstrate LENGTH() on student names.*

**Answer:**

```
SELECT Name, LENGTH(Name) AS NameLength FROM Student;
```

**Explanation:** LENGTH() returns the character length of string values.

### Question 14

*Question: Demonstrate CONCAT() using first and last names.*

**Answer:**

```
SELECT CONCAT('Aman', ' ', 'Sharma') AS FullName;
```

**Explanation:** CONCAT() combines multiple string arguments into a single string.

### Question 15

*Question: Demonstrate SUBSTRING()/SUBSTR() with suitable examples.*

**Answer:**

```
SELECT SUBSTRING('Database Management', 1, 8) AS SubStrVal;
```

**Explanation:** SUBSTRING(str, pos, len) extracts a portion of text starting at pos for length len.

## Section D: Conversion Functions

### Question 16

*Question: Convert a number into a character/string value.*

**Answer:**

```
SELECT CAST(12345 AS CHAR) AS StrValue;
```

**Explanation:** CAST() converts a numeric constant into string data type CHAR.

### Question 17

*Question: Convert a string into a numeric value.*

**Answer:**

```
SELECT CAST('987.65' AS DECIMAL(10,2)) AS NumValue;
```

**Explanation:** CAST() parses a numeric string into a DECIMAL number.

### Question 18

*Question: Convert a date value into a different format.*

**Answer:**

```
SELECT DATE_FORMAT(CURDATE(), '%d-%b-%Y') AS FormattedDate;
```

**Explanation:** DATE\_FORMAT() formats a date according to specifiers (e.g. 24-Sep-2026).

### Question 19

*Question: Demonstrate CAST() with suitable examples.*

**Answer:**

```
SELECT CAST('2026-09-24' AS DATE) AS ConvertedDate;
```

**Explanation:** Converts string literal into SQL DATE data type.

### Question 20

*Question: Demonstrate CONVERT() where supported by the DBMS.*

**Answer:**

```
SELECT CONVERT('100', SIGNED) AS SignedInt;
```

**Explanation:** CONVERT() converts an expression to a target data type in MySQL.

## Section E: Date Functions

### Question 21

*Question: Display the current date using a suitable date function.*

**Answer:**

```
SELECT CURDATE() AS TodayDate, CURRENT_DATE() AS CurrentDate;
```

**Explanation:** CURDATE() returns the current date from the operating system.

### Question 22

*Question: Extract the year, month and day from a date.*

**Answer:**

```
SELECT YEAR(CURDATE()) AS YR, MONTH(CURDATE()) AS MTH, DAY(CURDATE()) AS DY;
```

**Explanation:** YEAR(), MONTH(), and DAY() extract specific date components.

### Question 23

*Question: Find the difference between two dates.*

**Answer:**

```
SELECT DATEDIFF('2026-12-31', '2026-09-24') AS DaysDiff;
```

**Explanation:** DATEDIFF(expr1, expr2) calculates expr1 - expr2 in days.

### Question 24

*Question: Add a specified number of days to a date.*

**Answer:**

```
SELECT DATE_ADD('2026-09-24', INTERVAL 30 DAY) AS FutureDate;
```

**Explanation:** DATE\_ADD() adds an interval (30 days) to a starting date.

### Question 25

*Question: Display employees' joining dates in a required date format.*

**Answer:**

```
SELECT Name, DATE_FORMAT('2025-08-01', '%W, %M %d, %Y') AS FormattedJoining FROM  
Employee LIMIT 1;
```

**Explanation:** Applies DATE\_FORMAT() to display joining dates in verbose format.